MARIO MÉNDEZ

MACHINE LEARNING OPERATIONS ENGINEER (AI/ML) · SYSTEMS ARCHITECT

Production MLOps Lifecycle | LLM & Agentic Workflows | Model Serving Endpoints | Cloud Infrastructure & CI/CD

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Portfolio: morense-01.github.io/portafolio-profesional **| LinkedIn:** linkedin.com/in/mariomendeza **| GitHub:** github.com/morense-01

Google Skills Verified | Credly Badges (Cisco/IBM/AWS) | Microsoft Learn | Anthropic Claude Certified

PROFESSIONAL SUMMARY

Systems Engineer with a completed 4-year university degree and deep engineering acumen combining enterprise infrastructure reliability (99.9% uptime, NIST CSF security) with production **Machine Learning Operations (MLOps)**. Demonstrated track record deploying AI/ML models and Large Language Models (LLMs) into production environments via high-performance REST APIs and containerized microservice endpoints (Python, FastAPI, Docker). Experienced executing full MLOps lifecycles including experiment tracking (**MLflow**), data/model drift monitoring, automated CI/CD workflows (GitHub Actions), and scalable cloud data pipelines (BigQuery, Databricks). Advanced practitioner of **Agentic AI workflows (LangChain, LangGraph, RAG)** and modern AI-native developer tooling (Google Antigravity, Cursor, Windsurf, Claude Code, Gemini CLI). High ownership, forensic root-cause analysis, and seamless execution within distributed remote Agile/Scrum teams.

TECHNICAL ARSENAL & CORE COMPETENCIES

MLOps & Model Lifecycle: Model Serving (FastAPI, Docker endpoints), Experiment Tracking & Model Registry (MLflow), Model Monitoring & Drift Detection, Automated Retraining, CI/CD for ML (GitHub Actions), Cloud Run, AWS ECS/SageMaker.

Generative AI & Agentic Workflows: LLM Fine-tuning & API Integrations, Agentic Architectures (LangChain, LangGraph), Retrieval-Augmented Generation (RAG), Vector Embeddings (ChromaDB, FAISS, Pinecone), Prompt Engineering (Anthropic Certified).

AI-Enthusiastic Tooling & IDEs: Google Antigravity, Cursor, Windsurf, Claude Code CLI, Gemini CLI, Copilot CLI, OpenAI API, Anthropic Claude SDK.

Data Engineering & Cloud: Google Cloud Platform (Vertex AI, BigQuery, GKE), AWS (SageMaker, S3, CloudWatch), Microsoft Azure (Azure ML, AKS), Databricks, Apache Spark, SQL, Distributed Data Pipelines.

Programming & Core Tech: Python (FastAPI, Pydantic, Scikit-learn, PyTorch, Pandas, NumPy), Bash/Linux Shell, REST APIs, JSON/YAML, Git/GitHub.

Systems, Reliability & Security: Linux System Administration, Microservices, Containerization, Networking (TCP/IP, DNS, VPN), NIST CSF, Zero Trust, Root-Cause Forensics, SLA/SLO/SLI Governance, Extreme Ownership, Agile/Scrum.

PROFESSIONAL EXPERIENCE

AI/ML & Cloud Infrastructure Consultant · MLOps Lead | ODS Logistics (2024 – 2026 | Santo Domingo, DR (Remote & On-site))

- Engineered and deployed production AI/ML inference microservices and automated predictive pipelines using Python (FastAPI, Docker), sustaining 99.9% uptime for logistics operations across 30,000+ sq. ft. distribution hubs.
- Designed and productionized an internal Agentic Retrieval-Augmented Generation (RAG) assistant using LangChain, vector databases, and custom guardrails, accelerating compliance, customs, and operational data retrieval by 45%.
- Implemented end-to-end MLOps automation with GitHub Actions CI/CD: automated linting, test suites, container image builds, vulnerability scanning, and zero-downtime containerized deployments.
- Established model telemetry and observability: real-time monitoring of inference latency (p95/p99), input data schema validation with Pydantic, and automated alerts for prediction anomalies.
- Enforced NIST CSF cybersecurity standards and strict access control across model endpoints, APIs, and hybrid network interconnects.

Technology & Systems Engineering Manager | MADECO (2022 – 2024 | Santo Domingo, DR)

- Directed enterprise technology operations, hybrid cloud platforms (Azure/AWS), and mission-critical manufacturing execution systems.
- Architected automated telemetry and system anomaly detection pipelines in Python, improving proactive incident response by 40%.
- Led the migration of 63 legacy servers into virtualized and containerized environments, lowering operational and licensing costs by 18%.
- Automated data extraction and reporting pipelines connecting ERP systems (SAP Business One / S/4HANA) to central data storage via RESTful integrations.
- Led Agile cross-functional engineering teams, mentoring engineers in modern DevOps practices, forensic root-cause analysis, and SLA management.

Infrastructure & Systems Specialist | UNIPAGO (2020 – 2022 | Santo Domingo, DR)

- Managed high-throughput transaction processing infrastructure in a heavily regulated financial clearinghouse environment.
- Developed Python and Shell automation scripts for mission-critical batch processing, database consistency validation, and automated failover.
- Maintained 99.99% availability for critical financial services, ensuring compliance with strict data governance, auditing, and PCI-DSS standards.

Technology Manager · Industrial Systems | Dominicana de Cales (2018 – 2020 | Dominican Republic)

- Modernized computing, networking, and telemetry logging infrastructure supporting continuous industrial operations.
- Instituted structured incident management, automated system health checks, and continuous disaster recovery (BCP/DRP) procedures.

FEATURED AI/ML & MLOPS PROJECTS

Autonomous Agentic RAG Platform (LangGraph & FastAPI): Designed a production-grade multi-agent document reasoning platform utilizing LangGraph, LangChain, ChromaDB, and OpenAI/Claude APIs. Deployed containerized endpoints with dynamic context routing, hallucination guardrails, and automated evaluation metrics.

End-to-End MLOps Pipeline with MLflow & Docker: Implemented end-to-end regression and classification pipelines incorporating data preprocessing, automated hyperparameter tuning, experiment tracking with MLflow, container packaging in Docker, and automated CI/CD deployment to Google Cloud Run.

AI-Native Accelerated Engineering Workflows: Pioneered agentic coding workflows employing Google Antigravity, Cursor, Windsurf, Claude Code, and Gemini CLI for rapid microservice scaffolding, complex API integration, and automated test-driven development.

EDUCATION & ACADEMIC CREDENTIALS

Bachelor of Science in Computer Systems Engineering (Ingeniero en Sistemas de Computación)

4-Year College University Degree | Universidad Dominicana O&M;, Santo Domingo, Dominican Republic

Core Focus: Computer Architecture, Distributed Systems, Operating Systems, Software Engineering, Database Systems, Network Architecture.

VERIFIED CERTIFICATIONS & SPECIALIZED CREDENTIALS

Google Cloud Platform (16 Official Badges): Data, ML & AI Foundations · Cloud Computing Foundations · Infrastructure & Architecture · BigQuery Analysis · Generative AI & Large Language Models.

Anthropic Skilljar: Prompt Engineering with Anthropic Claude (Generative AI Architecture, Context Management & Model Evaluation).

Amazon Web Services (AWS): AWS Knowledge: Cloud Essentials / Architecture Fundamentals (Credly Verified).

IBM & Cisco Networking Academy: IBM Cybersecurity Fundamentals · IBM Cloud Essentials · Cisco CCNA (Enterprise Networking, Security & Automation).

Microsoft Learn & SAP: Microsoft Azure Administrator & Hybrid Cloud · SAP Enterprise Systems & Administration.